Solution: Smallest Number Range

Problem Statement

Given ‘M’ sorted arrays, find the smallest range that includes at least one number from each of the ‘M’ lists.

Example 1:

```
Input: L1=[1, 5, 8], L2=[4, 12], L3=[7, 8, 10]
Output: [4, 7]
Explanation: The range [4, 7] includes 5 from L1, 4 from L2 and 7 from
L3.
```

Example 2:

```
Input: L1=[1, 9], L2=[4, 12], L3=[7, 10, 16]
Output: [9, 12]
Explanation: The range [9, 12] includes 9 from L1, 12 from L2 and 10
from L3
```

Solution

This problem follows the **K-way merge pattern** and we can follow a similar approach as discussed in `Merge K Sorted Lists`.

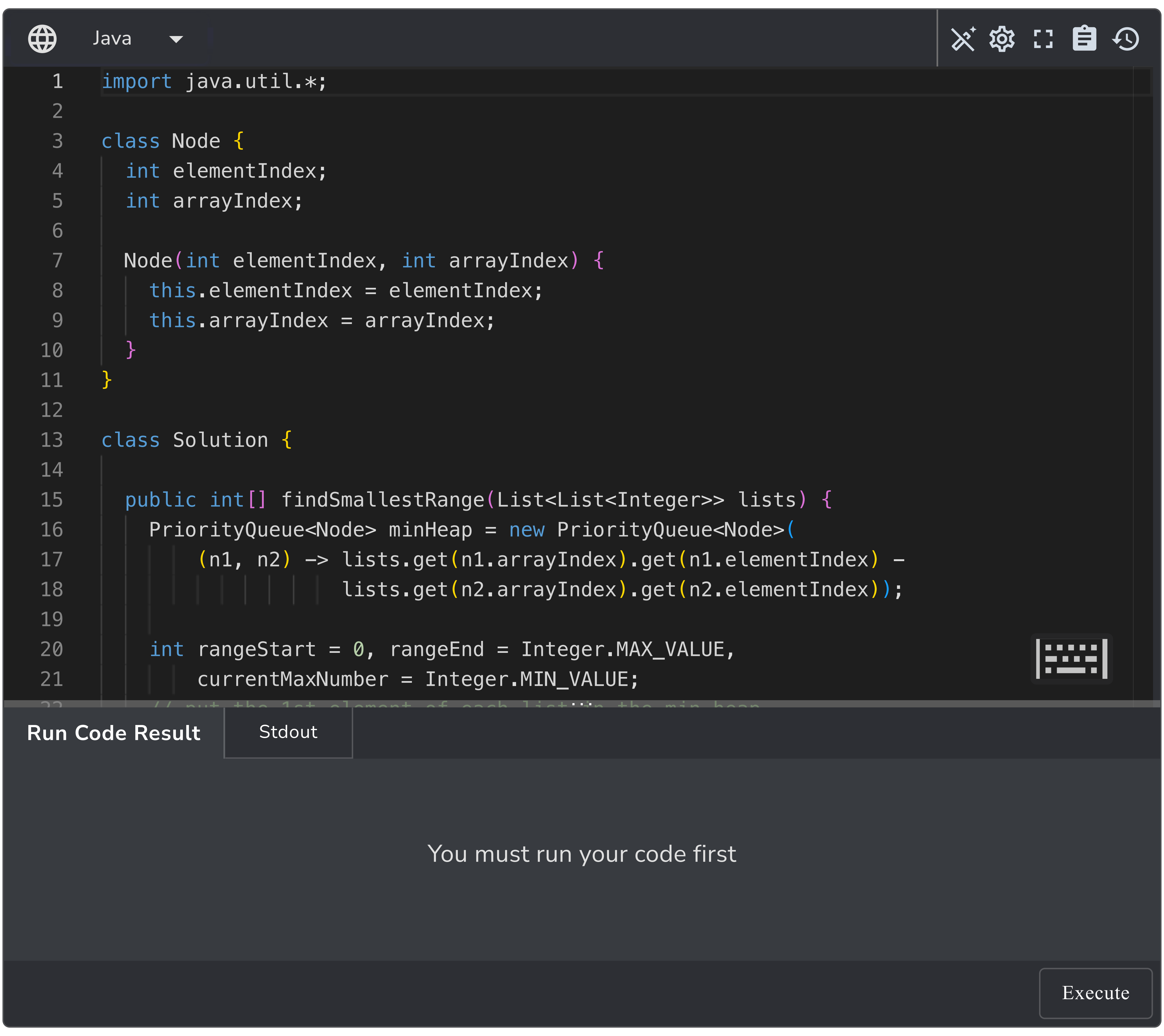
We can start by inserting the first number from all the arrays in a min-heap. We will keep track of the largest number that we have inserted in the heap (let’s call it `currentMaxNumber`).

In a loop, we’ll take the smallest (top) element from the min-heap and `currentMaxNumber` has the largest element that we inserted in the heap. If these two numbers give us a smaller range, we’ll update our range. Finally, if the array of the top element has more elements, we’ll insert the next element to the heap.

We can finish searching the minimum range as soon as an array is completed or, in other terms, the heap has less than ‘M’ elements.

Code

Here is what our algorithm will look like:



```java
import java.util.*;

class Node {
  int elementIndex;
  int arrayIndex;

  Node(int elementIndex, int arrayIndex) {
    this.elementIndex = elementIndex;
    this.arrayIndex = arrayIndex;
  }
}

class Solution {

  public int[] findSmallestRange(List<List<Integer>> lists) {
    PriorityQueue<Node> minHeap = new PriorityQueue<Node>(
        (n1, n2) -> lists.get(n1.arrayIndex).get(n1.elementIndex) -
                    lists.get(n2.arrayIndex).get(n2.elementIndex));

    int rangeStart = 0, rangeEnd = Integer.MAX_VALUE,
        currentMaxNumber = Integer.MIN_VALUE;
```

Time Complexity

Since, at most, we’ll be going through all the elements of all the arrays and will remove/add one element in the heap in each step, the time complexity of the above algorithm will be O(N*logM) where ‘N’ is the total number of elements in all the ‘M’ input arrays.

Space Complexity

The space complexity will be O(M) because, at any time, our min-heap will be store one number from all the ‘M’ input arrays.

← Previous
Smallest Number Range (Hard)

Next →
Problem Challenge 1: K Pairs With Largest Sums (Hard)

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